

Alan Turing

Alan Mathison Turing (/ˈtʃʊərɪŋ/; 23 June 1912 – 7 June 1954) was an English mathematician, computer scientist, logician, cryptanalyst, philosopher and theoretical biologist.^[6] He was highly influential in the development of theoretical computer science, providing a formalisation of the concepts of algorithm and computation with the Turing machine, which can be considered a model of a general-purpose computer.^{[7][8][9]} Turing is widely considered to be the father of theoretical computer science.^[10]

Born in London, Turing was raised in southern England. He graduated from King's College, Cambridge, and in 1938, earned a doctorate degree from Princeton University. During World War II, Turing worked for the Government Code and Cypher School at Bletchley Park, Britain's codebreaking centre that produced Ultra intelligence. He led Hut 8, the section responsible for German naval cryptanalysis. Turing devised techniques for speeding the breaking of German ciphers, including improvements to the pre-war Polish bomba method, an electromechanical machine that could find settings for the Enigma machine. He played a crucial role in cracking intercepted messages that enabled the Allies to defeat the Axis powers in the Battle of the Atlantic and other engagements.^{[11][12]}

After the war, Turing worked at the National Physical Laboratory, where he designed the Automatic Computing Engine, one of the first designs for a stored-program computer. In 1948, Turing joined Max Newman's Computing Machine Laboratory at the University of Manchester, where he contributed to the development of early Manchester computers^[13] and became interested in mathematical biology. Turing wrote on the chemical basis of morphogenesis^{[14][2]} and predicted oscillating chemical reactions such as the Belousov–Zhabotinsky reaction, first observed in the

Alan Turing

OBE FRS



Turing in 1951

Born	Alan Mathison Turing 23 June 1912 Maida Vale, London, England
Died	7 June 1954 (aged 41) Wilmslow, Cheshire, England
Cause of death	Suicide by cyanide poisoning ^[note 1]
Education	University of Cambridge (MA) Princeton University (PhD)
Known for	Cryptanalysis of the Enigma · Turing's proof · Turing machine · Turing test · unorganised machine · Turing pattern · Turing reduction · "The Chemical Basis of Morphogenesis" · Turing paradox
Partner	Joan Clarke (engaged in 1941 but did not marry)
Awards	Smith's Prize (1936)

daughter of Edward Waller Stoney, chief engineer of the Madras Railways. The Stoneys were a Protestant Anglo-Irish gentry family from both County Tipperary and County Longford, while Ethel herself had spent much of her childhood in County Clare.^[21] Julius and Ethel married on 1 October 1907 at the Church of Ireland St. Bartholomew's Church on Clyde Road in Ballsbridge, Dublin.^[22]

Julius's work with the ICS brought the family to British India, where his grandfather had been a general in the Bengal Army. However, both Julius and Ethel wanted their children to be brought up in Britain, so they moved to Maida Vale,^[23] London, where Alan Turing was born on 23 June 1912, as recorded by a blue plaque on the outside of the nursing home where he was born,^{[24][25]} which later became the Colonnade Hotel.^{[19][26]} Turing had an elder brother, John Ferrier Turing, father of Dermot Turing, 12th Baronet of the Turing baronets.^[27] In 1922, he discovered *Natural Wonders Every Child Should Know* by Edwin Tenney Brewster. He credited it with opening his eyes to science.^[28]

Turing's father's civil service commission was still active during Turing's childhood years, and his parents travelled between Hastings in the United Kingdom^[29] and India, leaving their two sons to stay with a retired Army couple. At Hastings, Turing stayed at Baston Lodge, Upper Maze Hill, St Leonards-on-Sea, now marked with a blue plaque.^[30] The plaque was unveiled on 23 June 2012, the centenary of Turing's birth.^[31]

Very early in life, Turing's parents purchased a house in Guildford in 1927, and Turing lived there during school holidays. The location is also marked with a blue plaque.^[32]

School

Turing's parents enrolled him at St Michael's, a primary school at 20 Charles Road, St Leonards-on-Sea, from the age of six to nine. The headmistress recognised his talent, noting that she "...had clever boys and hardworking boys, but Alan is a genius".^[33]

Between January 1922 and 1926, Turing was educated at Hazelhurst Preparatory School, an independent school in the village of Frant in Sussex (now East Sussex).^[34] In 1926, at the age of 13, he went on to Sherborne School,^[35] an independent boarding school in the market town of Sherborne in Dorset, where he boarded at Westcott House. The first day of term coincided with the 1926 General Strike, in Britain, but Turing was so determined to attend that he rode his bicycle unaccompanied 60 miles (97 km) from Southampton to Sherborne, stopping overnight at an inn.^[36]



English Heritage blue plaque in Maida Vale, London, marking Turing's birthplace in 1912



Turing at age 16, c. 1928 – c. 1929